CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Chandubhai S Patel Institute of Technology

Faculty of Technology and Engineering

Chandubhai S. Patel Institute of Technology (CSPIT)

Department of Computer Science & Engineering

Date: 07/10/2026

Laboratory Manul

Academic Year	:	2026-27	Semester	:	3rd
Course code	:	ITUE203	Course name	:	WEB DEVELOPMENT FRAMEWORKS

Practical – 11

Student Management CRUD with Search and Filter

Problem Statement:- Create a student management module to add, view, edit, delete, search and filter student records. All database operations must us MYSQLi statements and clear success/failure messages.

Objective:- To develop a Student Management Module using PHP and MySQLi that allows users to add, view, edit, delete, search, and filter student records efficiently, while performing all database operations using MySQLi statements and displaying clear success and failure messages for each operation.

db.php:

```
<?php

// Database connection
$host = "localhost";
$username = "root";
$password = "";
$database = "student_db";

$conn = mysqli_connect($host, $username, $password, $database);

if (!$conn) {
   die("Database Connection Failed: " . mysqli_connect_error());
}

mysqli_set_charset($conn, "utf8mb4");

?>
```

edit.php:

```
<?php

ob_start();

include 'db.php';

$error = "";

$id = isset($_GET['id'])
   ? (int) $_GET['id']
   : 0;

if ($id <= 0) {
   die("Invalid student ID.");
}

/* Fetch student record */

$stmt = mysqli_prepare(
   $conn,
   "SELECT id, name, email, course, gender
    FROM students
    WHERE id = ?"
);

if (!$stmt) {
   die(
      "Database statement error: " .
      mysqli_error($conn)
   );
}

mysqli_stmt_bind_param(
   $stmt,
   "i",
```

```
    $id
);

mysqli_stmt_execute($stmt);

$result = mysqli_stmt_get_result($stmt);

if (
    !$result ||
    mysqli_num_rows($result) == 0
) {
    die("Student record not found.");
}

$student = mysqli_fetch_assoc($result);

mysqli_stmt_close($stmt);

/* Update student record */

if (isset($_POST['update_student'])) {

    $name = trim($_POST['name']);
    $email = trim($_POST['email']);
    $course = $_POST['course'];
    $gender = $_POST['gender'];

    $stmt = mysqli_prepare(
        $conn,
        "UPDATE students
         SET name = ?,
             email = ?,
             course = ?,
             gender = ?
         WHERE id = ?"
    );

    if (!$stmt) {

        $error =
            "Database statement error: " .
            mysqli_error($conn);

    } else {

        mysqli_stmt_bind_param(
            $stmt,
            "ssssi",
            $name,
            $email,
            $course,
            $gender,
            $id
        );

        if (mysqli_stmt_execute($stmt)) {

            mysqli_stmt_close($stmt);
```

```
            header(
                "Location: index.php?message=Student updated successfully"
            );

            exit();

        } else {

            $error =
                "Failed to update student: " .
                mysqli_stmt_error($stmt);

            mysqli_stmt_close($stmt);
        }
    }
}

?>
```

index.php:

```
<?php

ob_start();

include 'db.php';

$message = "";
$error = "";

// Display success message
if (isset($_GET['message'])) {
    $message = $_GET['message'];
}

/* Add student */

if (isset($_POST['add_student'])) {

    $name = trim($_POST['name']);
    $email = trim($_POST['email']);
    $course = $_POST['course'];
    $gender = $_POST['gender'];

    $stmt = mysqli_prepare(
        $conn,
        "INSERT INTO students (name, email, course, gender)
         VALUES (?, ?, ?, ?)"
    );

    if ($stmt) {

        mysqli_stmt_bind_param(
            $stmt,
            "ssss",
            $name,
            $email,
```

```
            $course,
            $gender
        );

        if (mysqli_stmt_execute($stmt)) {

            mysqli_stmt_close($stmt);

            header(
                "Location: index.php?message=Student added successfully"
            );

            exit();

        } else {

            $error = "Failed to add student: " .
                    mysqli_stmt_error($stmt);

            mysqli_stmt_close($stmt);
        }

    } else {

        $error = "Database statement error: " .
                mysqli_error($conn);
    }
}

/* Delete student */

if (isset($_GET['delete'])) {

    $id = (int) $_GET['delete'];

    if ($id > 0) {

        $stmt = mysqli_prepare(
            $conn,
            "DELETE FROM students WHERE id = ?"
        );

        if ($stmt) {

            mysqli_stmt_bind_param(
                $stmt,
                "i",
                $id
            );

            if (mysqli_stmt_execute($stmt)) {

                mysqli_stmt_close($stmt);

                header(
                    "Location: index.php?message=Student deleted successfully"
                );
```

```
            exit();

        } else {

            $error = "Failed to delete student: " .
                mysqli_stmt_error($stmt);

            mysqli_stmt_close($stmt);
        }

    } else {

        $error = "Database statement error: " .
            mysqli_error($conn);
    }

  } else {

    $error = "Invalid student ID.";
  }
}

/* Search and filter */

$search = isset($_GET['search'])
  ? trim($_GET['search'])
  : "";

$course_filter = isset($_GET['course_filter'])
  ? $_GET['course_filter']
  : "";

/* Fetch student records */

if ($search != "" && $course_filter != "") {

  $search_value = "%" . $search . "%";

  $stmt = mysqli_prepare(
    $conn,
    "SELECT id, name, email, course, gender
     FROM students
     WHERE (name LIKE ? OR email LIKE ?)
     AND course = ?
     ORDER BY id ASC"
  );

  mysqli_stmt_bind_param(
    $stmt,
    "sss",
    $search_value,
    $search_value,
    $course_filter
  );

} elseif ($search != "") {

  $search_value = "%" . $search . "%";
```

```
    $stmt = mysqli_prepare(
        $conn,
        "SELECT id, name, email, course, gender
         FROM students
         WHERE name LIKE ? OR email LIKE ?
         ORDER BY id ASC"
    );

    mysqli_stmt_bind_param(
        $stmt,
        "ss",
        $search_value,
        $search_value
    );

} elseif ($course_filter != "") {

    $stmt = mysqli_prepare(
        $conn,
        "SELECT id, name, email, course, gender
         FROM students
         WHERE course = ?
         ORDER BY id ASC"
    );

    mysqli_stmt_bind_param(
        $stmt,
        "s",
        $course_filter
    );

} else {

    $stmt = mysqli_prepare(
        $conn,
        "SELECT id, name, email, course, gender
         FROM students
         ORDER BY id ASC"
    );
}

if ($stmt) {

    if (mysqli_stmt_execute($stmt)) {

        $result = mysqli_stmt_get_result($stmt);

    } else {

        $error = "Failed to fetch students: " .
                 mysqli_stmt_error($stmt);

        $result = false;
    }

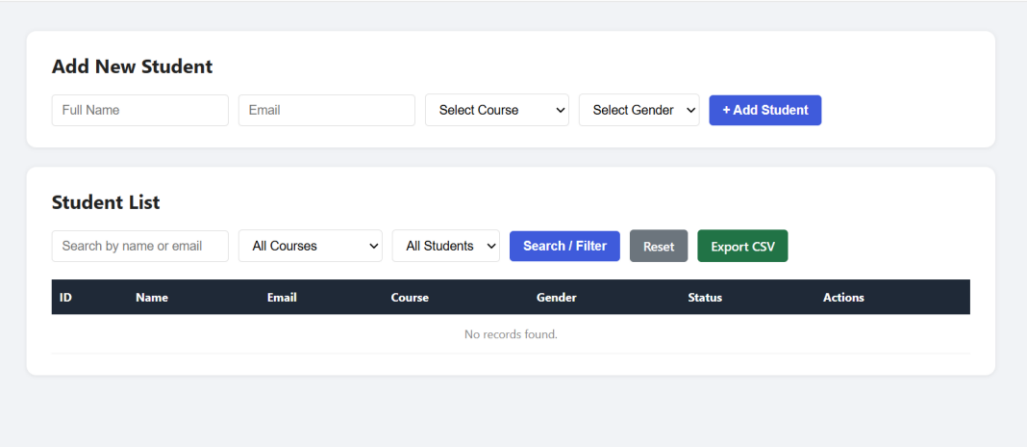
} else {
```

```
    $error = "Database statement error: " .
        mysqli_error($conn);

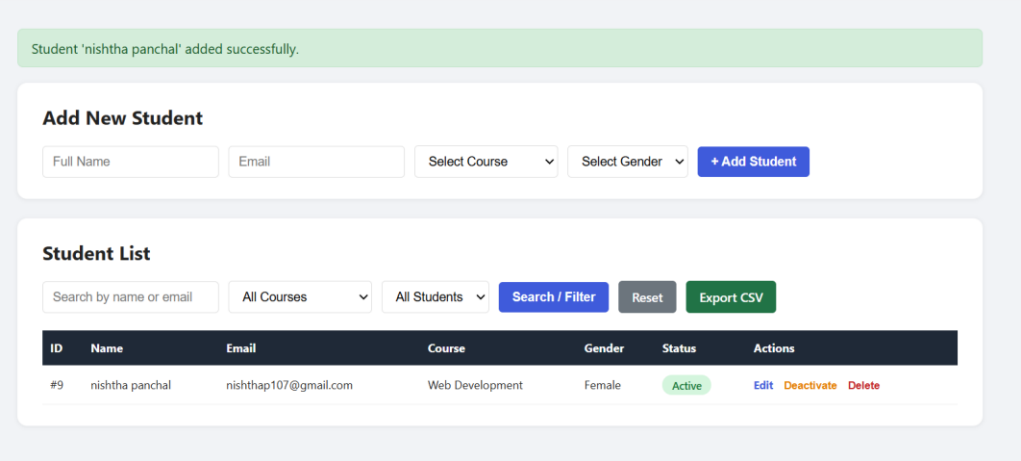
    $result = false;
}

?>
```

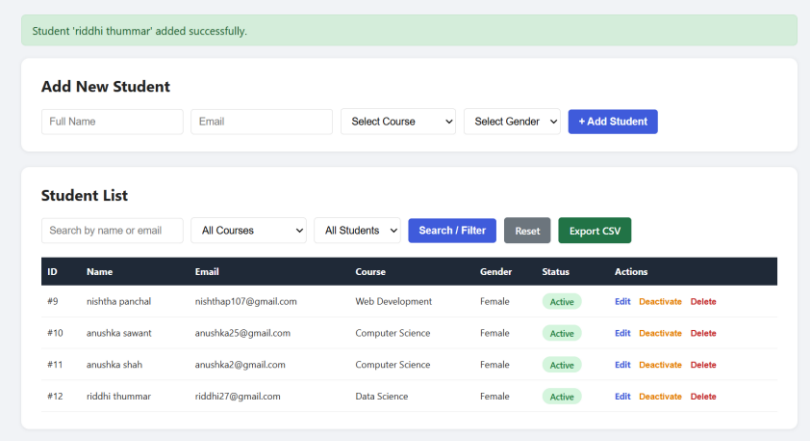
Student Management System form:



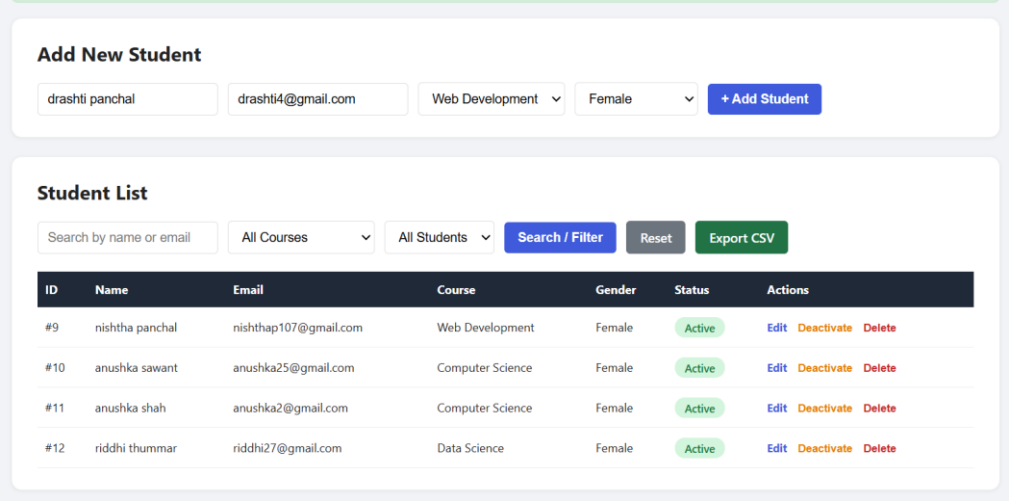
Successful Student Registration:

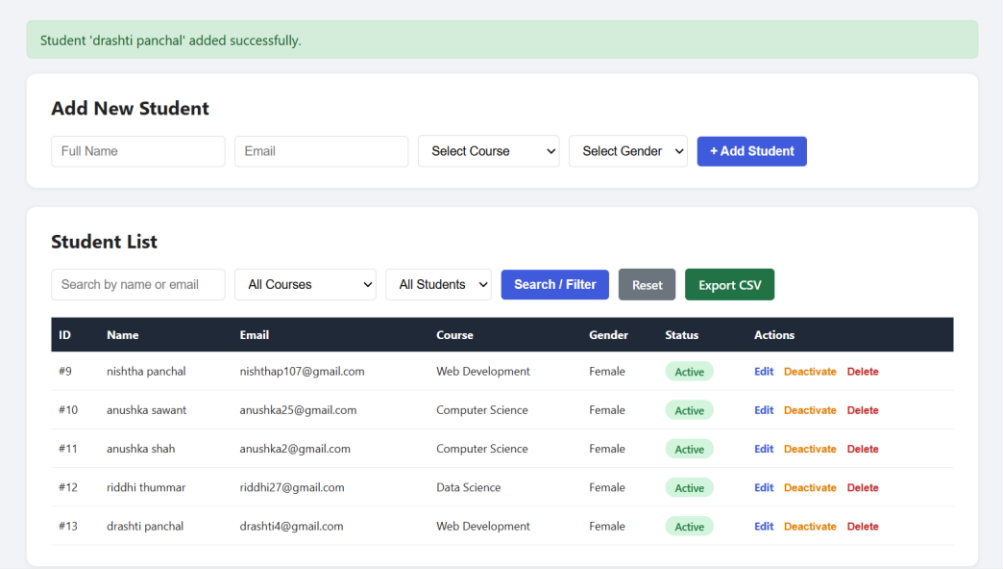


Student records displayed in a table:

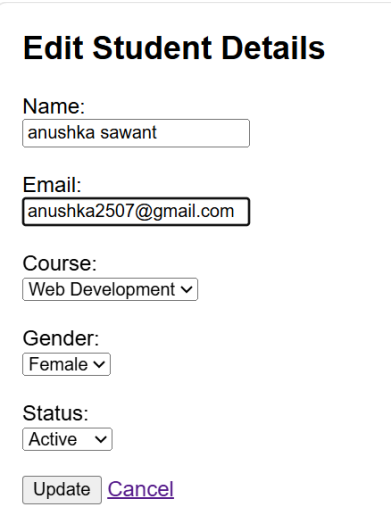


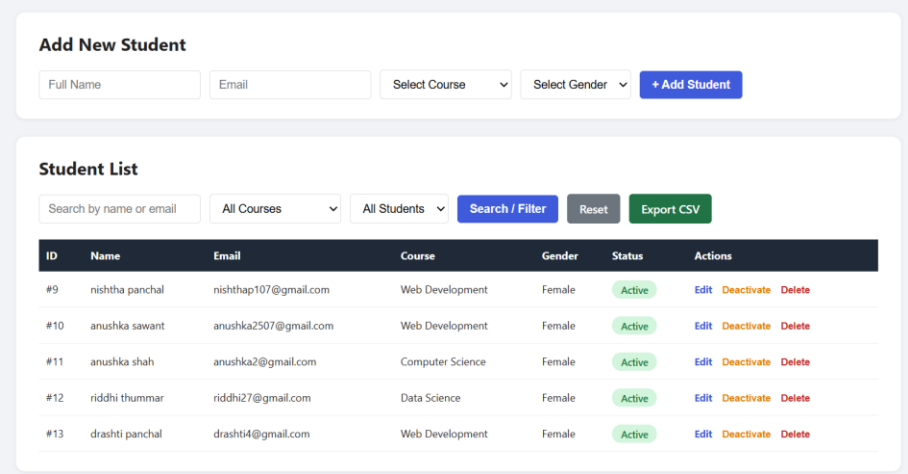
Successful addition of a new student:



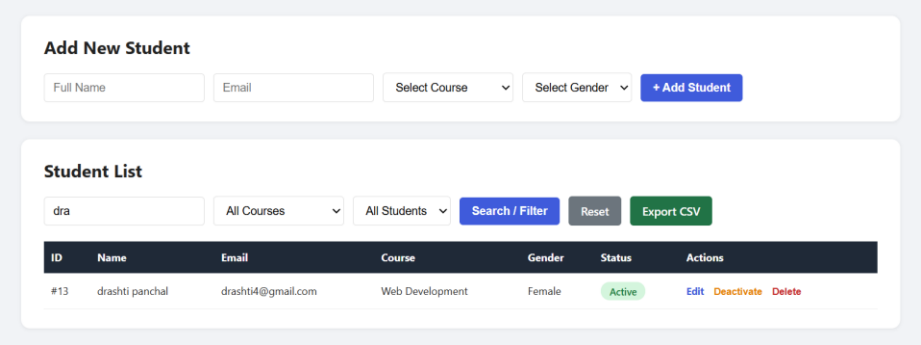


Successful update of student details:

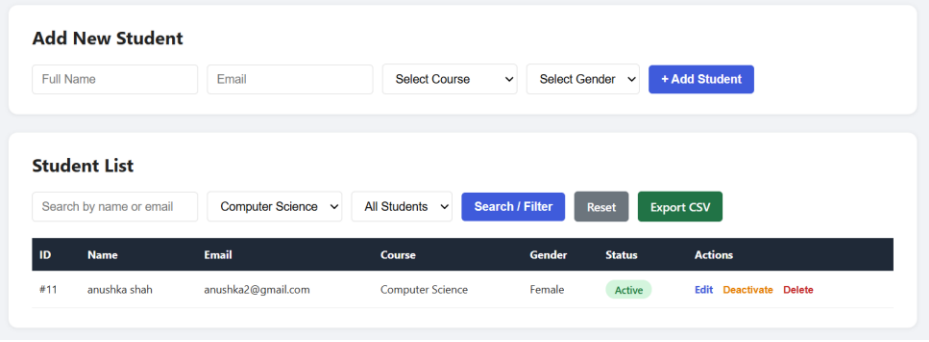




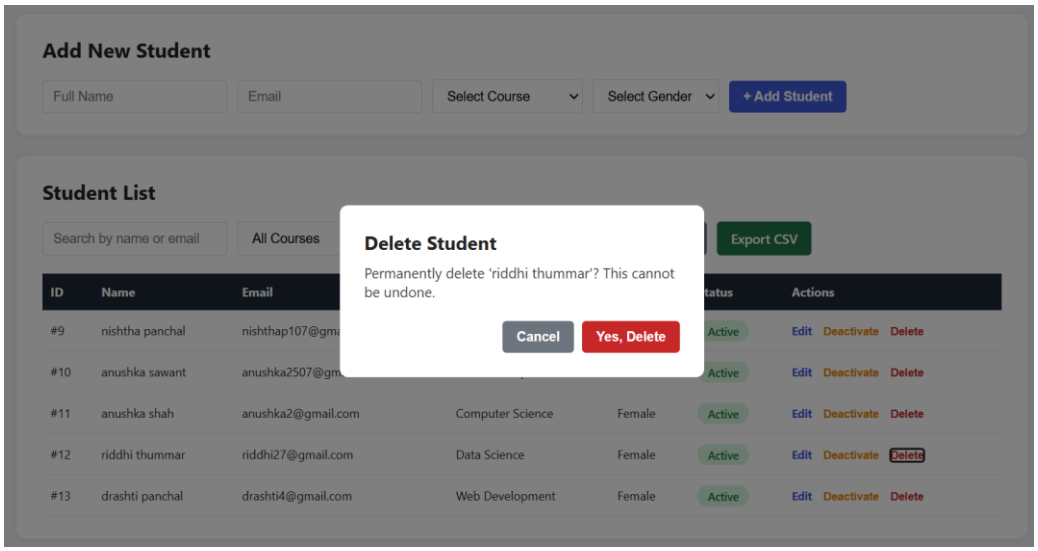
Search functionality:

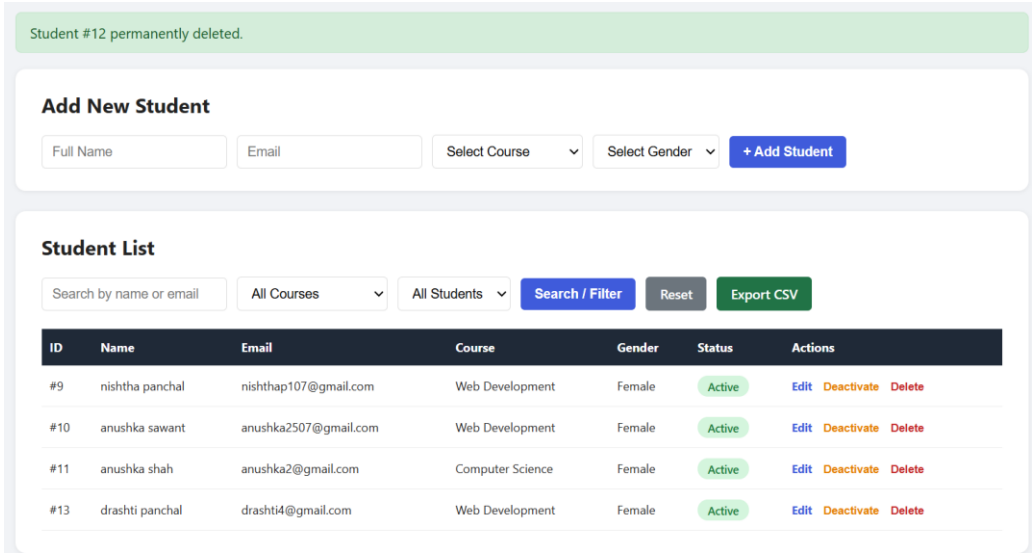


Course-wise filtering:

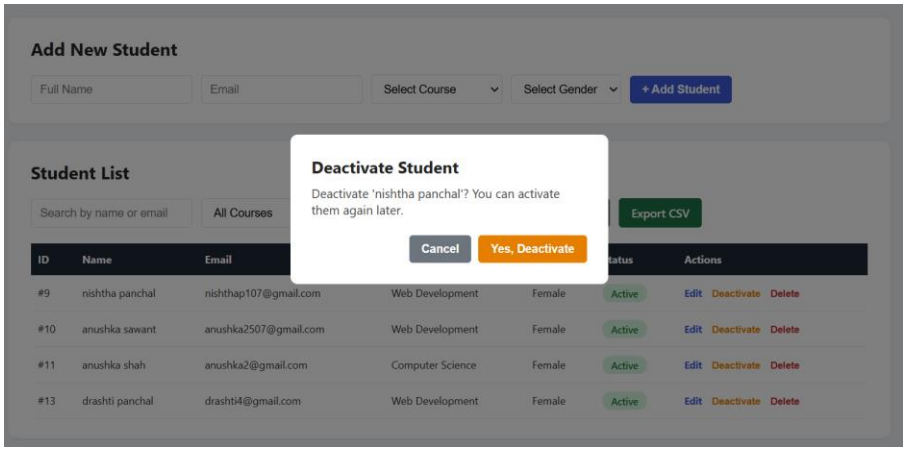


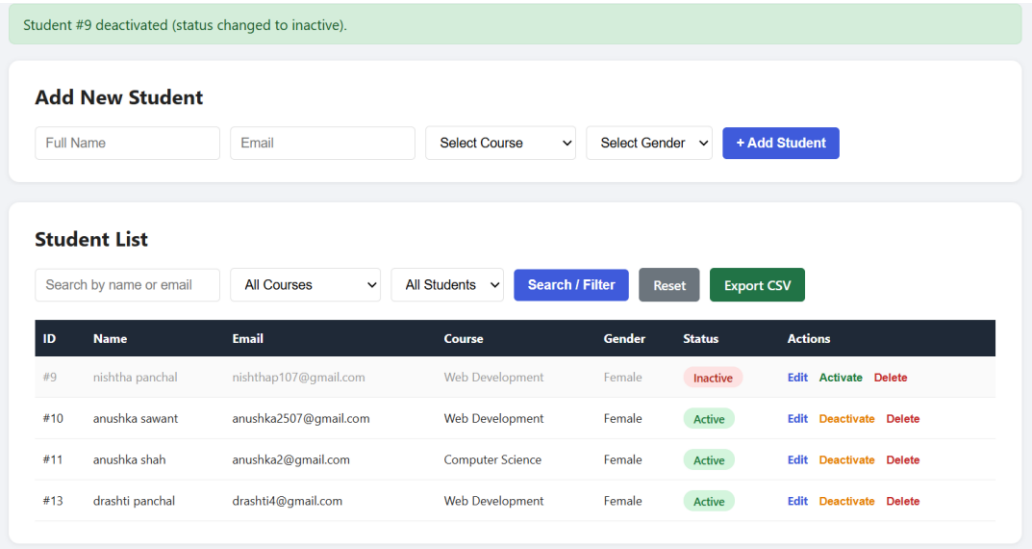
Delete confirmation message:





Successful soft delete (student status changed to inactive):





Exported CSV file opened in Excel:

	A	B	C	D	E	F
	ID	Name	Email	Course	Gender	Status
	13	drashti panchal	drashti4@gmail.com	Web Development	Female	active
	11	anushka shah	anushka2@gmail.com	Computer Science	Female	active
	10	anushka sawant	anushka2507@gmail.com	Web Development	Female	active
	9	nishtha panchal	nishthap107@gmail.com	Web Development	Female	inactive

Learning Outcome:

To learn how to develop and manage a Student Management Module using PHP and MySQLi, perform CRUD operations (Create, Read, Update, and Delete), implement search and filter functionality, use MySQLi prepared statements for secure database operations, and display clear success or failure messages for each operation.